

Programming Exercise 1

Using the IDE you installed on your computer, or using the school VM -

Write the following Python program using your Python IDLE shell by clicking the File tab → New File.

Then run and test it. When they build and run correctly, copy the code into a separate text file for each program, with the same name as that program. *Do this by copying the program's code into Notepad (or TextEdit for Macs) and saving it as a .txt file.*

Submit the text files by opening the eCampus **Programming Exercise 1** assignment by clicking on its title.

Go to the "Assignment Submission" section of the web page and click the "Browse Local Files" button.

Attach the text files for all 3 programs and click the "Submit" button at the bottom of the page.

NOTE: All programs that you write must have 3 lines with comments at the very top with 1) the program name, 2) your name, and 3) a sentence describing what the program does.

1. Write a new program called **Program1A**.
 - a. Copy the following code into your new IDLE window. Make sure all lines of code are left aligned with no spaces in front of them. Replace the instruction on the second line with your name.

```
# Program1A
# Write your name here
# Purpose: This program calculates the area of a rectangle
width = int(input("Enter width: "))
height = int(input("Enter height: "))
area = width * height
print("The area is ", area, " square units.")
```

- b. Run the program and enter a width of 2 and length of 5 when asked; it should print a statement saying that the area is 10.
2. Start a new program and name it **Program1B**.
 - a. Using the above program as a pattern, change the code to calculate the area of a triangle by following these instructions:
 - i. Ask the user to enter the base of the triangle and put the answer in a variable named **base**. (Like you asked for the width in the program above.)
 - ii. Ask the user to enter the height of the triangle and put the answer in a variable named **height**. (Like you asked for the height in the program above.)
 - iii. Set **area** equal to $.5 * \text{base} * \text{height}$
 - iv. Copy the print statement from the program above and paste it on the next line.
 - b. An example of the program input and output is shown below:
Enter base: 10
Enter height: 20
The area is 100 square units
 - c. Run and test your program to make sure it works before submitting it.
3. Write a new program named **Program1C**.

- a. Ask the user to enter an integer and put it into a variable called **length**.
- b. Ask the user to enter another integer and put it into a variable called **width**.
- c. Set **area** equal to **length * width**
- d. Set **cost** equal to **area * 2.50**
- e. Print the area and cost with the following format:
A rug with **area** square feet is **cost** dollars.
(Where **area** and **cost** are the variables you just calculated.)

An example of the program input and output is shown below:

Enter length: 20

Enter width: 15

A rug with 300 square feet is 750 dollars

- f. Run and test your program to make sure it works before submitting it.